

Computational Physics and Data Science Exercise 3 - Linear Regression by Methods of Varying Efficacy

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This exercise analysed a portion of the GAIA probe's dataset using clustering techniques and principal component analysis, along with standard scaling techniques, to produce a replica Hertzsprung-Russel diagram for the data. After disregarding the majority of the data as uninteresting background objects, a small coherent group of data points with similar measurements for the twelve features of the data frame were extracted and used for the majority of the exercise. Key results were the similarities and differences of the Hertzsprung-Russel diagram for this data compared to a standard diagram, the differences in efficiency for different clustering techniques and the principal components' highest contributing features for the first and second principal components.

Part 1

Plotting Proper Motion Characteristics

The comma separated file was read using pandas and cleaned for invalid data points such as 'NaN' entries or blank spaces. Then the proper motion in right ascension (pmra) and declination (pmdec) were plotted against each other to get a handle on the shape of the data, as seen in figure 1 below. The number of data points before filtering was 97606.

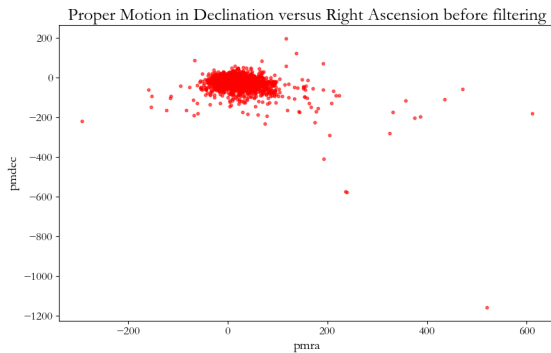


Figure 1: A plot of proper motion characteristics before filtering the data.

A pair of distribution plots were then generated to view the frequency of data points at each value of pmra and pmdec. The grid lines were set to be one standard deviation of each feature apart, such that a rough estimate of how many standard deviations away from the mean the outliers were could be taken. This was found to be about 2 deviation away on the close side of the main spike in the data, and about 5 away on the far side. The data frame was then filtered using a number of conditions such as: parallaxes must be greater than 0, relative error on parallax must be no bigger than 20%, pmdec was between -60 and 15 and pmra was between -14 and 26 (values taken from figure 2).

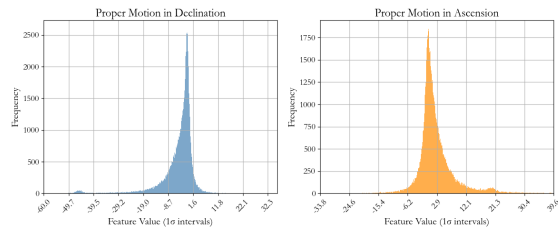


Figure 2: Distribution plots for proper motion in declination and right ascension with vertical grid lines representing numbers of standard deviations from the mean.

Figure 3 now shows the data post-filtering, focused on the main clusters of the available data. The number of data points after filtering was 25300, so 72306 data points were excluded by the filtering.

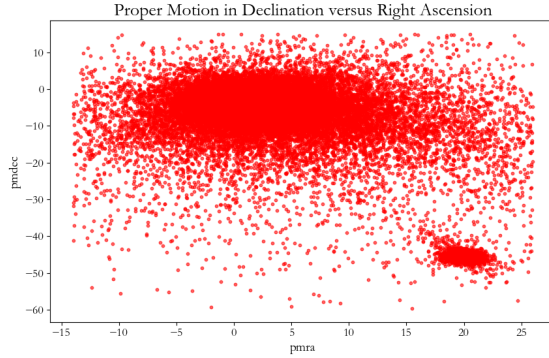


Figure 3: A plot of proper motion characteristics after filtering the data using the limits determined from the above distribution plots.

An additional column, relative parallax error, was added to the data frame to hold values for this feature, as filtering was performed using bounds in relative error as opposed to absolute error.

Part 2

Feature	Mean Value
ra	56.686288
dec	24.449604
l	166.277122
b	-23.306837
parallax	1.563091
pmra	4.041285
pmdec	-7.757551
g	16.273475
bp	17.035493
rp	15.449017
parallax error	0.098856
ruwe	1.130673
relative parallax error	8.535170

The objective of part 2 was to practice scaling the data to a standard size, such that they all

had a standard deviation of 1, thus putting their actual values on the same standard axes, which proves useful for the principal component analysis in part 4. Above are the mean values for each feature, and below are their respective standard deviations.

Feature	Mean Value
ra	1.097783
dec	0.991934
l	1.092871
b	0.987730
parallax	1.585925
pmra	7.160520
pmdec	11.005215
g	1.921847
bp	2.196929
rp	1.776955
parallax error	0.102433
ruwe	0.796668
relative parallax error	5.704145

After scaling, the features' mean values were as follows:

Feature	Mean Value
ra	$6.466220e - 15$
dec	$-1.163830e - 15$
l	$2.022997e - 14$
b	$-2.884860e - 15$
parallax	$3.089316e - 17$
pmra	$-3.033147e - 17$
pmdec	$-2.836554e - 17$
g	$-1.436672e - 15$
bp	$4.587635e - 16$
rp	$-5.135988e - 16$
parallax error	$-7.007131e - 17$
ruwe	$5.925870e - 17$
relative parallax error	$2.224308e - 16$

As can be predicted by the logic of standard scaling, the standard variation for each feature post-scaling was 1.00002 (the 2 hundred-thousandth arising from inherent python uncertainties).

Part 3

K-means and Gaussian Mixture Clustering

By taking note of the number of stars in the interesting cluster as `n_components/n_clusters` was increased for each clustering technique, it was possible to obtain the minimum number of clusters for which the interesting stars data frame contained around 1000 data points.

The K-means style clustering proved unsuccessful in clustering together less than 1200 stars without surpassing 20 clusters, which seemed excessive. Past 10 clusters, the change in number of interesting stars for each additional cluster was minimal, so the `n_clusters` hyperparameter was left at 10. Due to the nature of the Voronoi cells into which this technique partitions the data, it can be seen in figure 4 that additional clusters would only serve to divide up the larger chunk of the data, leaving only one cluster for the smaller block and thus having minimal impact on the number of interesting stars. For 10 clusters as seen in figure 4, the number of these such stars was 1257.

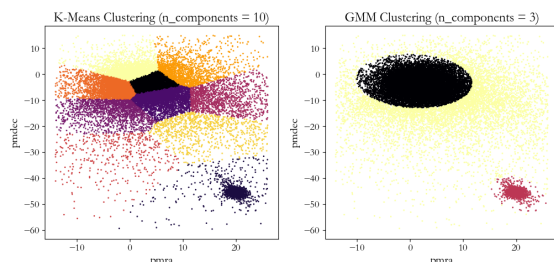


Figure 4: K-Means and Gaussian Clustering `pmra` vs `pmdec` plots with the number of components set to the smallest number required for the important cluster (bottom left) to contain around 1000 stars.

For the Gaussian Mixture Modelling (GMM) style, as predicted, a trivial case of 2 clusters was unsuccessful for capturing around 1000 stars in the smaller bunch, however an `n_components` value of 3 was perfect, achieving a total of 1082 stars in the `unscaled_interesting_stars` data frame (unscaled being the recommended way to leave

the data before performing clustering). As for why 2 clusters may have failed, GMM assumes that each cluster is Gaussian-distributed, so for clusters with big differences in relative size compared to the total amount of data (with the large cluster having 99% and the small having 1% of the total data), it seems that the algorithm sacrificed efficient clustering of the smaller chunk of points for the sake of the larger chunk, thus grouping the smaller chunk into the secondary, outer cluster as seen in figure 5. For this case, the number of stars in the secondary cluster was around 5000 - 5 times larger than the expected number.

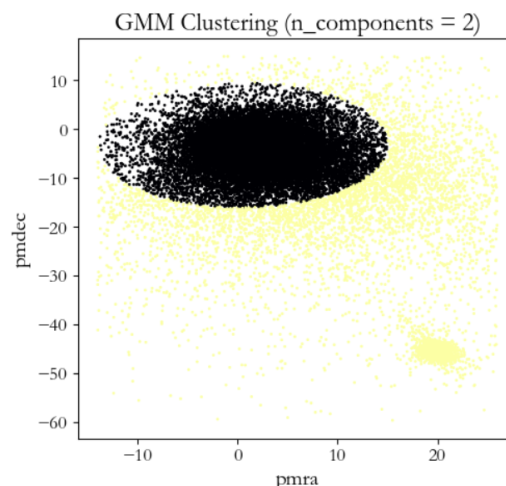


Figure 5: GMM clustering with `n_components = 2`.

Spectral Clustering

As an extension, an extra clustering method, spectral clustering, was performed for comparison with K-means and GMM techniques. The closest the number of interesting stars got to 1000 was 939 at an `n_clusters` of 21. In terms of efficiency, this method falls far behind K-means and GMM, which both managed to get close to 1000 with less than half the number of components. however, in terms of proximity of 1000 stars in the interesting stars cluster, this method wins as it had only 61 data points too few, while GMM had 82 too many.

It was noted that as the number of components for this method increased, the decrease in the number of stars in the interesting cluster decayed significantly similar to K-means, except not as severely. In this respect, spectral clustering was more useful as the number of interesting stars did not get stuck around 1200, rather it kept decreasing as more clusters were added.

The reason for the relative failure of spectral clustering compared to GMM may have been similar to that of K-means, as both techniques rely on constructing a similarity graph and using eigenvalue decomposition. Given that most of the data was grouped into one cluster, most of these eigenvalues will have been dominated by this cluster, which explains why adding clusters merely divided up the main chunk of data, although the number of interesting stars decreased to a satisfactory value as a consequence.

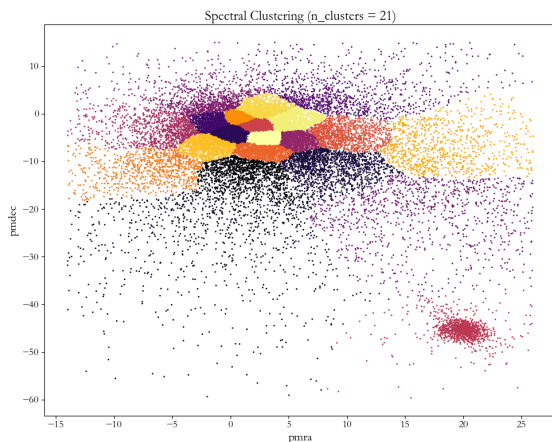


Figure 6: Spectral clustering performed for the same data with 21 clusters, which was the number required to bring the number of stars in the interesting cluster (bottom right) as close to 1000 as possible.

Part 4

Covariance Matrix

Part 4 began with standardising the data, as was done in part 2 for practice.

Below is the covariance matrix for the twelve features of the data set, hinting at the linear trends between each feature. Given that positive values imply positive correlation and vice versa for negative values, the most strongly proportional features were red photometric band magnitude and G-band magnitude, which makes sense physically as these are similar forms of measurement.

The greatest inverse proportionality was between galactic longitude and declination. This shows that the Milky Way bulges towards its centre when viewed from Earth, becoming thinner at its edges where centrifugal force counteracts gravitational forces to prevent matter from collapsing towards the centre, as has happened at the central bulge of the galaxy as gravity has dominated, drawing matter together despite its angular momentum.

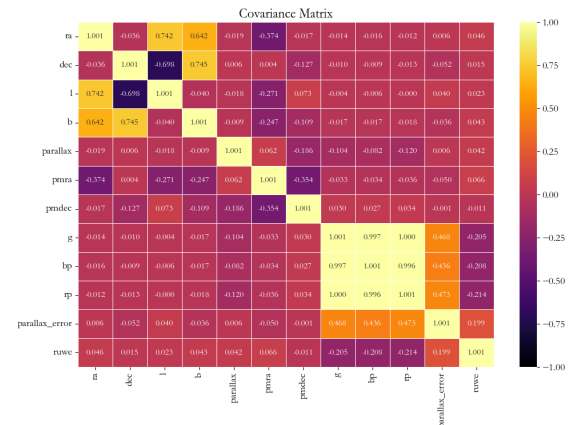


Figure 7: The covariance matrix for each feature of the data set.

Extracting Eigenvectors and Eigenvalues

In solving the diagonalisation problem for this principal component analysis, a heat map for the eigenvectors of each principal component and each feature was created. The highest modular eigenvector values in each row shows the features which contribute most to that principal

component, so for PC1 these were G-band magnitude, blue photometric and red photometric band magnitude with shared values of -0.54. for PC2, these features were right ascension, proper motion in right ascension and galactic longitude. These findings were confirmed in part 5, where the loadings of each PC were found.

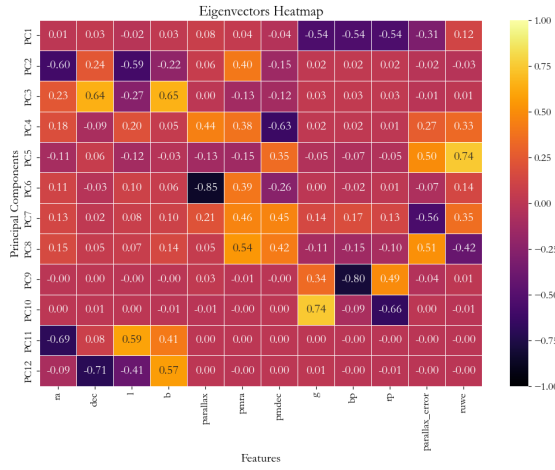


Figure 8: An eigenvector heat map for each principal component and feature of the data.

In figure 9, the eigenvalues of each principal component (otherwise known as the explained variance for each) is shown as a bar chart. For PC1 and PC2, these values were 3.32844514 and 2.23058968 respectively. These eigenvalues were used for further analysis in the next section.

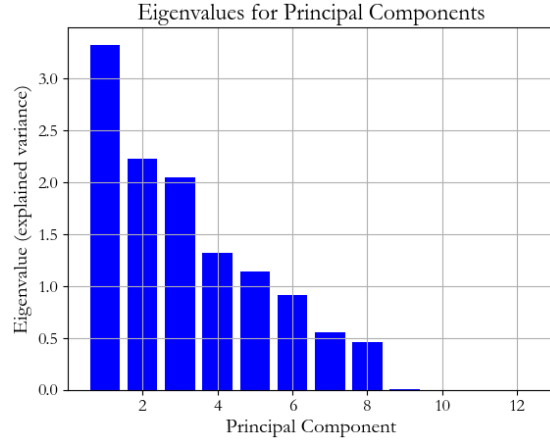


Figure 9: The explained variance/eigenvalues for each principle component.

PC1 and PC2

By analysing the explained variance ratios for each principle component, it was determined that PC1 and PC2 contribute 27.71% and 18.57% of the total variance respectively, for a combined total of 45.86% variance accountability. The explained variance ratios for each other principal component are shown in the code, but not included here as PC1 and PC2 were the main interests of this analysis.

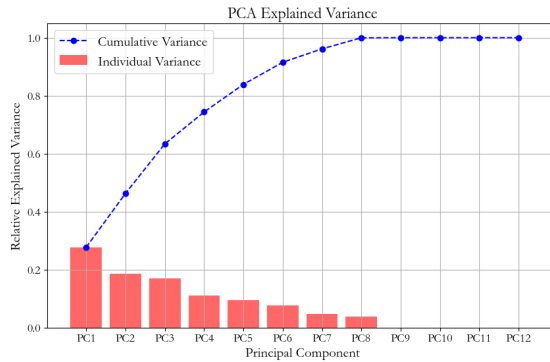


Figure 10: A combined scatter plot and bar chart displaying the individual and cumulative explained variance percentages for each principal component.

Part 5

Projection of Principal Components

The difference between PC1 and PC2 in explained variance ratio is only 9.14%, explaining why the projection of the two components shows no clear trend as shown in figure 11. This is in part due to the scaling applied to the data. In order to further analyse these components, their loadings were found.

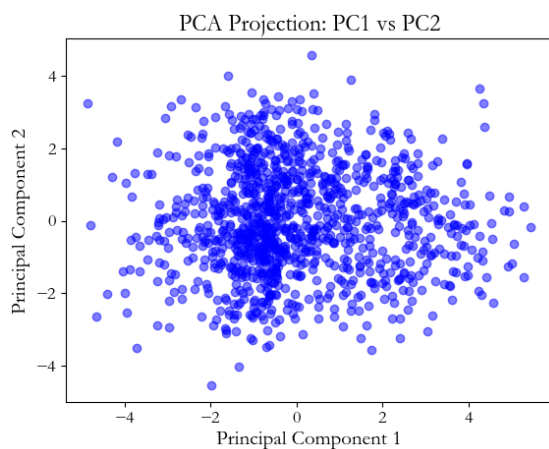


Figure 11: The projection of PC1 over PC2.

Loadings of PC1 and PC2

The loadings of PC1, otherwise known as the highest contributing features to this component, were the red photometric band magnitude, G-band magnitude and blue photometric magnitude, as follows in the table, with the eigenvector values for each loading being the modulus of the eigenvector value from the eigenvector heat map in figure 8.

Loading (PC1)	Eigenvector value
rp	0.542745
g	0.541703
bp	0.537687

The loadings for PC2 were right ascension, proper motion in right ascension and galactic longitude, as shown in the table below.

Loading (PC2)	Eigenvector value
ra	0.598207
l	0.589264
pmra	0.401218

These results are interpreted to mean that the first principal component, contributing the most to the variance of the data set, was dominated by measurements of the intensity of light from each part of the electromagnetic spectrum for the stars observed. This makes sense given that the variety of distances, redshifts and amount of drifting matter that could affect measurements of light from a star in the Milky Way vary greatly and randomly per star.

The factors contributing to PC2 were associated with position and velocity of stars as measured along the Milky Way's plane and the celestial equator. Given that speed of a star can be affected by position relative to the centre of the galaxy, mass and local solar system configuration among other factors, it makes sense that measurements such as these would contribute greatly to the variance of the data set.

Hertzsprung-Russel Diagram

Figure 12 shows a graph with absolute magnitude on the ordinate and colour index on the abscissa, resembling a Hertzsprung-Russel diagram. The colour scale has been chosen to liken the shape of the diagram to other examples, where stars are classified into different categories based on their evolutionary stage. An example of such a diagram is shown in figure 13.

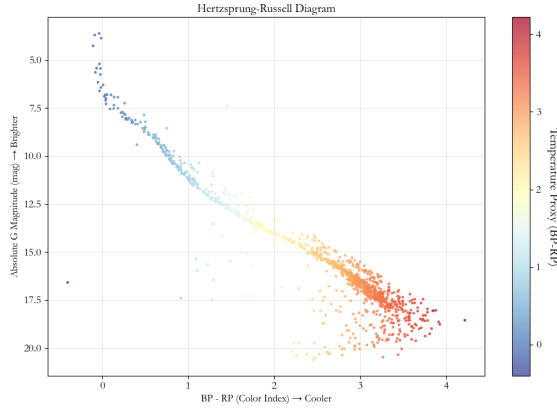


Figure 12: A Hertzprung-Russell-like diagram constructed using the loadings of PC1.

From the differences between these two figures, we can see that the Milky Way mostly consists of main sequence stars, with very few if any giants and white dwarfs.

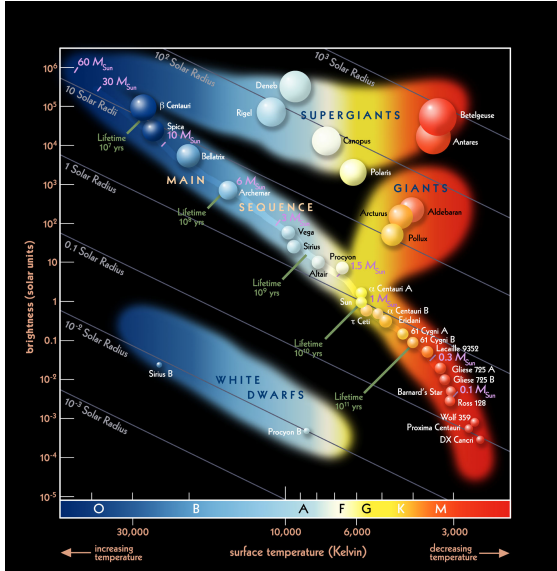


Figure 13: An example Hertzprung-Russell diagram.

Figures 14, 15 and 16 show three different angles of the 3D plot created with different axes for bp and rp.

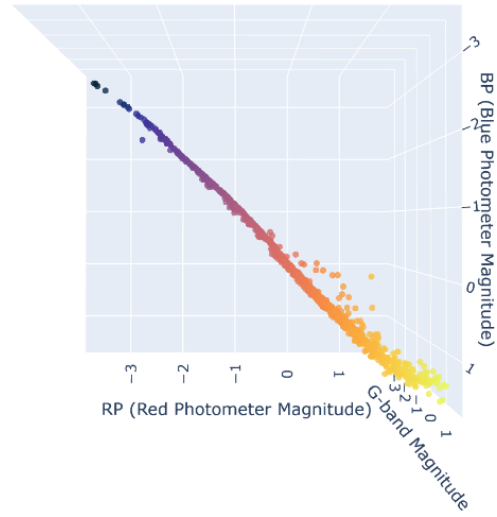


Figure 14: A graph of blue photometric band magnitude against red photometric band magnitude.

Figure 14 shows an inverse relationship between blue and red photometric band magnitudes for the data set, implying that most stars have a mix towards the middle values of these two measurements, with few existing at either extreme.

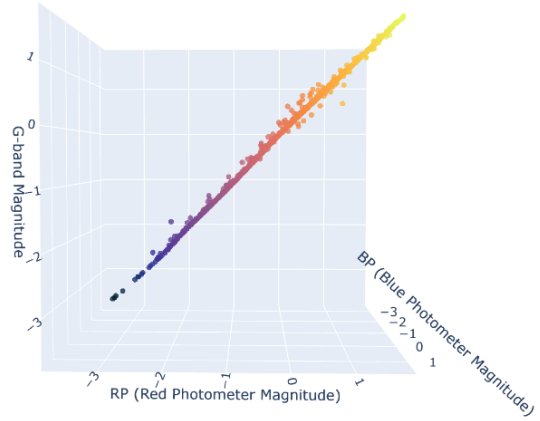


Figure 15: A graph of absolute magnitude against red photometric band magnitude.

Figure 15 shows a linear relationship between rp and G-band magnitude, implying that redder

stars tend to be brighter in general, with some variation at rp values around 0.

Figure 16: A graph of absolute magnitude against blue photometric band magnitude.

Figure 16 shows a relationship similar to that of figure 14, implying that bluer stars tend to be dimmer, which ties into ideas of redshift affecting the number of stars seen towards the upper end of the electromagnetic spectrum.

